FIITJEE Students From All Programs have bagged 34 in Top 100, 66 in Top 200 and 174 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2014: 2521 FIITJEE Students from Classroom / Integrated School Programs & 3579 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2014.

ALL INDIA INTEGRATED TEST SERIES

FIITJEE

JEE(Main)-2015

ANSWERS, HINTS & SOLUTIONS

PART TEST–III
(Main)

Q. No.	PHYSICS	CHEMISTRY	MATHEMATICS
1.	D	C	C
2.	D	B	C
3.	C	D	C
4.	B	A	A
5.	B	C	C
6.	A	B	B
7.	A	A	D
8.	C	A	B
9.	C	C	A
10.	A	C	C
11.	C	C	A
12.	B	D	A
13.	C	A	A
14.	A	C	C
15.	C	C	D
16.	D	D	D
17.	D	A	D
18.	A	A	A
19.	A	D	C
20.	A	C	A
21.	A	A	A
22.	A	D	A
23.	B	C	A
24.	C	B	C
25.	B	A	B
26.	D	A	B
27.	D	A	C
28.	B	B	B
29.	A	C	D
30.	D	C	C

Physics

PART – I

SECTION – A

1. $4\times\frac{\mu_0 I}{4\pi}.\left[\sin 45^0+\sin 45^0\right]\left[\frac{1}{R}-\frac{1}{2R}\right]=\frac{\mu_0 I}{\pi}\left[\sqrt{2}\right]\left[\frac{1}{2R}\right]=\frac{\mu_0 I}{\sqrt{2}\pi R}$

2. $V_1=\frac{10}{4+4}\times 4$ volts = 5 volt

$$V_2=\frac{10\times\frac{100\times 4}{100+4}}{4+\frac{100\times 4}{100+4}}$$

= 4.9 volts $\therefore |V_1-V_2|=0.1$ volt

3. In steady state :-

$V_P-V_Q=\frac{E}{2R}\times R=\frac{E}{2}$

$V_Q-V_S=\frac{E}{2R}\times R=\frac{E}{2}$

$V_S=V_R$

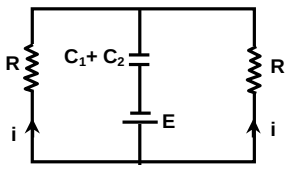
$V_P-V_R=E$

$\therefore Q_A=CE$

$Q_B=\frac{CE}{2}$

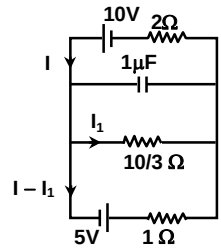
4. $V_P-V_0=-\left[10\times 10+10\times 10\right]$

= - 200 volts.

5. 

6. $\frac{10}{3}I_1+2I-10=0-5+I-I_1+2I-10=0$

After solving we get, $I_1=0$, Q = 0



7. $V=\frac{1}{4\pi\varepsilon_0\times 1}\left[q+\frac{q}{2}+\frac{q}{4}+\frac{q}{8}+\ldots\right]=\frac{q}{4\pi\varepsilon_0}\left[\frac{1}{1-(1/2)}\right]=\frac{q}{2\pi\varepsilon_0}$

8. Field at any internal point of the cavity.

$$\vec{E} = G\frac{4}{3}\pi\rho\left(\overrightarrow{C_2C_1}\right)$$

Since C_1 and C_2 are fixed (cavity being concentric with C_2 all the time), field remains constant both in magnitude and direction.

9. If the fifth point P has identical charges the field at the centre of symmetry O would be zero. Hence the field due to one single such charge at P would produce a field $\vec{E}'$, such that $(\vec{E}' + \vec{E}) = 0$

$$\therefore \quad \vec{E}' = -\vec{E} = -(-3\hat{i} + 4\hat{j} - 2\hat{k})$$

this field will be four times $\vec{E}'$ due to this rearrangement, hence the required field at O is

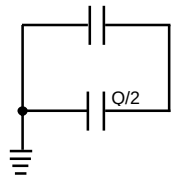
$$4\vec{E}' = (12\hat{i} - 16\hat{j} + 8\hat{k}) \text{ units}$$

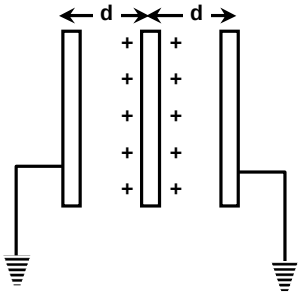
10. If $I_G = 0$ then it becomes a balanced whetstone bridge

$$\ell_1 = \frac{2}{3}\times 1\text{m} = \frac{2}{3}\times 100 = \frac{200}{3}\text{cm}$$

12. $$E_1 = \frac{\sigma}{\epsilon_0} = \frac{Q}{2A\epsilon_0}$$

$$V = E_1 d = \frac{Qd}{2A\epsilon_0}$$





13. Apply gaus theorem

$$E\left(4\pi r^2\right) = \frac{q_{in}}{\epsilon_0} \Rightarrow E = \frac{q}{4\pi\epsilon_0 r^2} \Rightarrow E \propto \frac{1}{r^2} \Rightarrow \text{field depends only on r.}$$

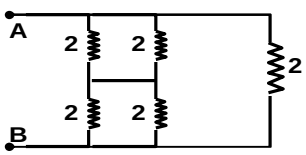
14. $$\frac{\Delta R}{R} = \frac{\Delta V}{V} + \frac{\Delta I}{I}$$

$$\Rightarrow \frac{\Delta R}{R}\times 100 = \frac{\Delta V}{V}\times 100 + \frac{\Delta I}{I}\times 100$$

$$= 3\%$$

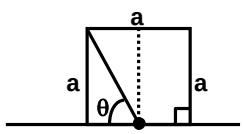
17. Magnetic field is zero at the centre due to symmetrical current distribution.

18.



19. $$\text{B at O} = \frac{2\times\mu_0(I/4)}{4\pi(a/2)}\left(0 + \frac{2}{\sqrt{5}}\right) + \frac{\mu_0(I/4)}{4\pi a}\left(\frac{2}{\sqrt{5}} + \frac{2}{\sqrt{5}}\right)$$

$$= \frac{2\mu_0 I}{4\sqrt{5}\pi a} + \frac{\mu_0 I}{4\pi a\sqrt{5}} = \frac{3\mu_0 I}{4\sqrt{5}\pi a}$$



20. Due to electrostatic shielding, the ratio will be 1.

21. When key is open
$q_1 = CE$
when key is closed
$$q_2 = C\left[\frac{E}{R_1 + R_2}R_2\right]$$
$$\therefore \frac{q_1}{q_2} = \left[\frac{CE}{CER_2}(R_1 + R_2)\right] = \frac{(R_2 + R_1)}{R_2} = \frac{5}{3}$$

22. If I_1 is current in ACDB then
$$\frac{\mu_0}{4}\cdot\frac{i_1}{r} = \frac{\mu_0}{4}\cdot\frac{i_2}{4r}$$
$$\frac{i_1}{i_2} = \frac{1}{4}$$
$$i_1 = \frac{1}{5}\times i = \frac{i}{5}$$

23. From symmetry the currents in segment gh is zero. Thus, the equivalent resistance R is given by
$$\frac{1}{R} = \frac{1}{4} + \frac{1}{4} + \frac{1}{2} + \frac{1}{2} = \frac{3}{2} \Rightarrow R = \frac{2}{3}\Omega$$
Current in eh = $\frac{10}{4} = 2.5A$
$\therefore$ Potential difference across eh = 5V

24. It is a balanced Wheatstone bridge for capacitor. Consequently the potentials of points a and b are equal. Thus energy stored in the capacitor is zero.

25. Let after steady state the charges on the capacitor be q_1 and q_2
$q_1 = CV$, $q_2 = CV$ where v is the potential difference between the plates.
Also from charge conservation
$q_1 + q_2 = 0$
$\Rightarrow$ V = 0
Thus, final energy stored in the capacitors $= \frac{1}{2}CV^2 + \frac{1}{2}CV^2 = 0$
Initial energy stored = $\frac{q^2}{2C} + \frac{q^2}{2C} = \frac{q^2}{C}$
Thus, heat developed in the resistor = q^2/C

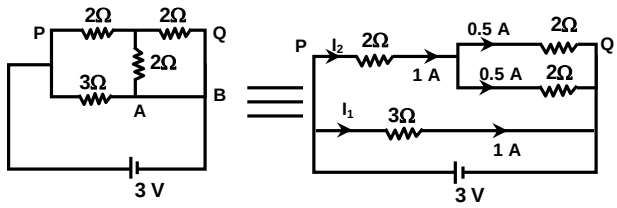
26. Initial capacitor is charged to Q/2
Energy stored in capacitor = $\frac{1}{2}\left(\frac{Q}{2}\right)^2 \times \frac{1}{C} = \frac{1}{8}\frac{Q^2}{C} \Rightarrow \frac{1}{8}CE^2 = U_i$
Finally energy stored in the capacitor, $U_f = \frac{1}{2}CE^2$

Work done by the battery = $\left(\frac{3}{2}CE\right)\times E = \frac{3}{2}CE^2$

$\therefore$ Heat energy lost = $\frac{3}{2}CE^2 - (U_f - U_i) = \frac{3}{2}CE^2 - \left(\frac{1}{2}CE^2 - \frac{1}{8}CE^2\right)$

$= CE^2\left(\frac{3}{2} - \frac{1}{2} + \frac{1}{8}\right) = \frac{9}{8}CE^2$

27. The equivalent circuit is
Current flowing through 3Ω resistance = 3/3 = 1 Amp
Current flowing through P to Q is also 1 amp.
From the figure current flowing through AB is 1 + 0.5 = 1.5 Amp



28. By using the relation

$B = \frac{\mu_0 i\theta}{4\pi r}$ where θ is in radian

Here $\theta = 180°$ i.e. π

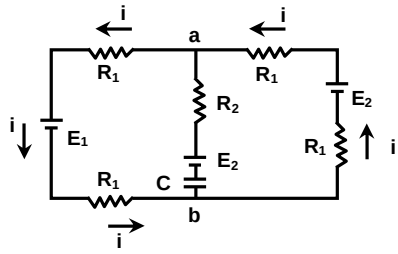
So, $B_p = \frac{\mu_0 I}{4r}$

29. Let x = a′ potential is zero (region II)

$k\frac{q}{a'} = k\frac{2q}{a + a'}$

$\Rightarrow$ a′ = a
So, in the region II,
Potential is zero and can be proved.
A solution can also found in region I (x = a/3).

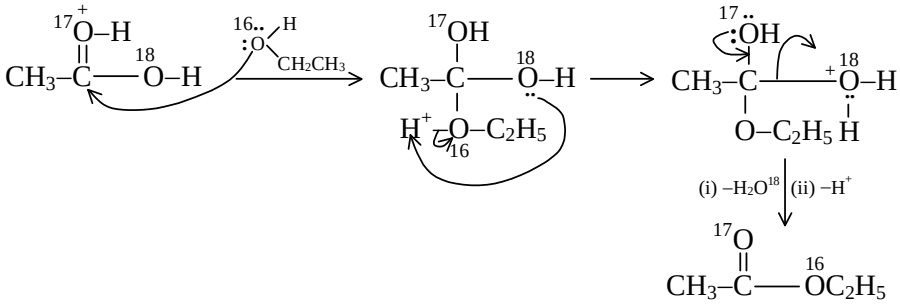
30. In the steady state
$E_2 - iR_1 - iR_1 - E_1 - iR_1 - iR_1 = 0$
$E_2 - E_1 = 4iR_1$
$I = \frac{E_2 - E_2}{4R_1} = (6.3 - 2.1)/(4\times 1.7) = \frac{2.1}{3.4}$
Solving, we get, $V_a - V_b = -iR_1 + E_2 - iR_1$
$= 6.3 - 2\times 1.7\times\frac{2.1}{3.4} = 6.3 - 2.1 = 4.2$ V

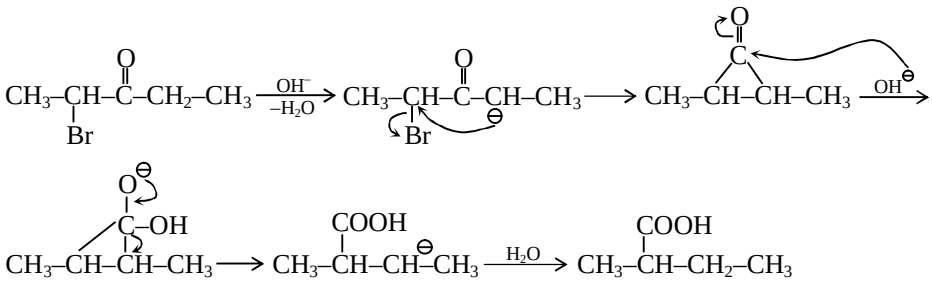


Chemistry

PART – II

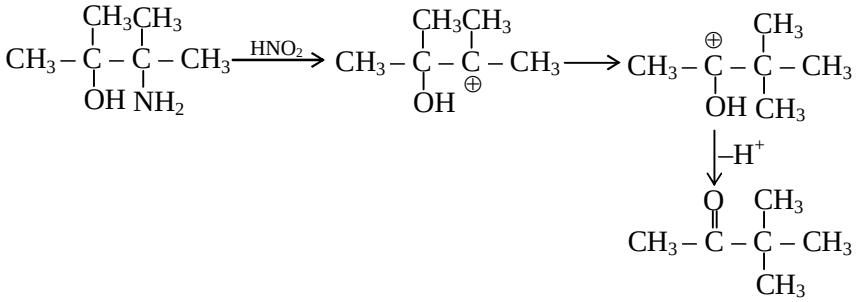
SECTION – A

1. 

3. 

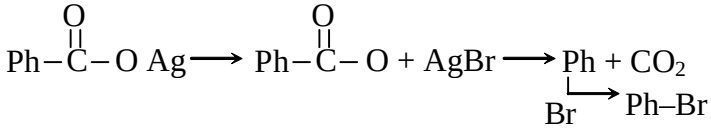
4. This is Baltz Schiemann reaction.

5 $4C_2H_5–Cl + NH_3 \longrightarrow \left[\begin{matrix} & C_2H_5 & \\ C_2H_5– & N^+ & –C_2H_5 \\ & C_2H_5 & \end{matrix}\right] Cl^-$

6. 

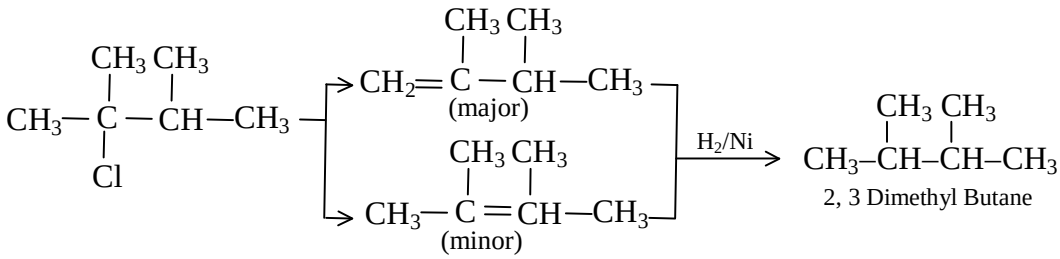
7. Carbyl amine test is responded by only 1° aliphatic/aromatic amines.

8. This is Hundsdicker reaction.



9. $(C_6H_5)_3CCl$ is hydrolysed most rapidly by SN_1 because $(C_6H_5)_3C^+$ is most stable.

10. SN_1 reaction is accompanied by racemisation while SN_2 reaction is accompanied by inversion of configuration.

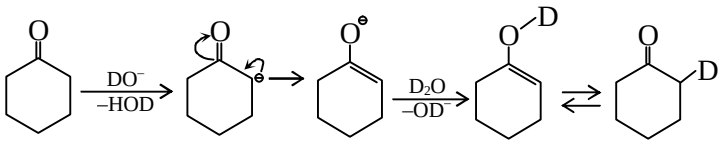
11. 

12. Nucleophilic addition.

13. Chloroform is formed when acetone is mixed with bleaching powder ($CaOCl_2$).

14. Due to intramolecular H–bonding.

15. Because $–CH_2–$ hydrogen is more acidic than $–CH_3$ hydrogen.

16. 

These steps are continued unless all the four α-H's are replaced by D.

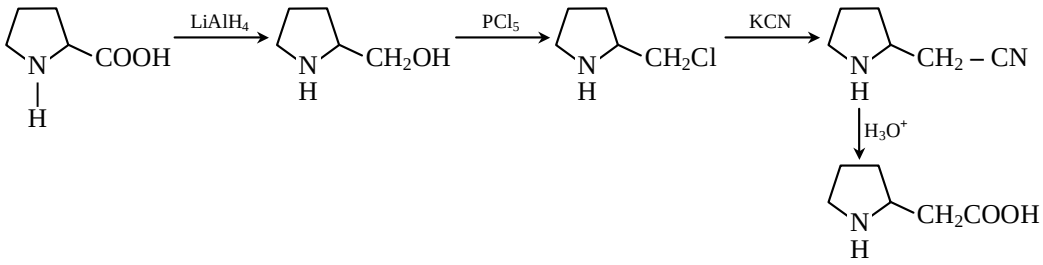
17. $CH_3–CH_2–\overset{O}{\overset{\|}{C}}–CH_3$ contain $–\overset{O}{\overset{\|}{C}}–$ and keto methyl group both.

18. In polystyrene the monomeric unit is styrene which is having the formula of $H_2C=CH—C_6H_5$.

19. Step growth polymers (or condensation polymers) are prepared by reaction between two functional groups and thereby eliminating small molecules during polymerization. Nylon-66 is an example of step growth polymer.

20. Test of terminal alkynes.

21. Compound 'A' give positive test with Na metal. That means it contains polar Hydrogen i.e., alcoholic group. It gives negative test with neutral $FeCl_3$ implies it does not contain of a phenolic group. It gives positive Lucas test means it is an alcohol.

22 

23. $RCOOH + CH_2N_2 \longrightarrow RCOOCH_3 + N_2$; methyl esters are formed.

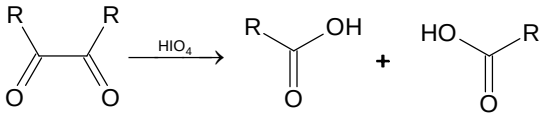
24. '2' position is attached with two electron withdrawing groups.

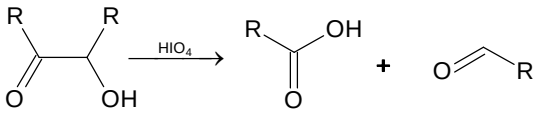
25. $LiAlH_4$ can reduce both ketone and ester.

27. n CH_2=CH–CH=CH_2 + n C_6H_5–CH=CH_2 (styrene)
(1,3-butadiene)

$\longrightarrow$ [CH_2–CH=CH–CH_2–CH(C_6H_5)–CH_2]$_n$ —
(Buna–S)

28. $CH_3NH_2 \xrightarrow{CH_3I\ (excess)} (CH_3)_3\overset{+}{N}I^- \xrightarrow{AgOH} (CH_3)_4\overset{+}{N}-OH^- \xrightarrow{\Delta} (CH_3)_3N + CH_3OH$

30. (1) 

(2) 

(4) $R-CHOH-CHOH-R \xrightarrow{HIO_4} R-CH=O + R-CH=O$

Mathematics PART – III

SECTION – A

1. $\int \frac{1-7\cos^2 x}{\sin^7 x \cos^2 x}dx = \int \frac{\sec^2 x}{\sin^7 x}dx - \int \frac{7}{\sin^7 x} = \frac{\tan x}{\sin^7 x} + c$ S

2. $\tan x \int \tan^{-1}\alpha \, d\alpha = \tan x\left[\alpha \tan^{-1}\alpha - \int \frac{\alpha}{1+\alpha^2}d\alpha\right] = \tan x\left\{\alpha \tan^{-1}\alpha - \frac{1}{2}\ell n\left(1+\alpha^2\right)\right\} + c$

3. $k = \int_0^1 \frac{e^t}{1+t}\,dt = \int_0^1 e^t \ell n(1+t)\,dt = \left[\ell n(1+t)e^t\right]_0^1 - \int_0^1 \frac{e^t}{1+t}dt = e\ell n2 - k$

4. $I_n = \int_0^\pi e^x (\sin x)^n\,dx = e^x(\sin x)^n\Big|_0^\pi - n\int_0^\pi (\sin x)^{n-1}\cos x\, e^x dx$

   $I_n = \frac{n(n-1)I_{n-2}}{1+n^2} \Rightarrow \frac{I_n}{I_{n-2}} = \frac{n(n-1)}{1+n^2}$

5. $\int_{-2k\pi}^{2k\pi} |\sin x|[\sin x]\,dx = \int_0^{2k\pi} |\sin x|\left([\sin x] + [-\sin x]\right)dx$

   $= -2k\int_0^\pi \sin x\,dx = \left[-2k(-\cos x)\right]_0^\pi = -4k$

   $I_k = -4k$

   $\sum_{k=1}^{100} I_k = -20200$

6. $f(n) - nf(n-1) = (-1)[f(n-1) - (n-1)f(n-2)] = (-1)^{n-1}[f(1) - f(0)] = (-1)^n$

   $\therefore \frac{f(n)}{n!} = \frac{f(n-1)}{(n-1)!} + \frac{(-1)^n}{n!}$ and this gives

   $f(n) = n!\left[1 - \frac{1}{1!} + \frac{1}{2!} - ..... + \frac{(-1)^n}{n!}\right]$

8. $x_n = \frac{1}{n}(n \in N) \to 0$ as $n \to \infty$ by the definition of continuity $f(x_n) \to f(0)$ as $n \to \infty$. But $f(x_n) = 0$ for all $x \in N$, so $f(0) = 0$

   Also $\lim_{x\to 0} \frac{f(x)-f(0)}{x-0} = \lim_{x\to 0} \frac{f(x_n)-f(0)}{x_n - 0}$ $\because$ f is derivable at x = 0

9. Equation of tangent $\frac{y - 2a\sin\theta}{x - a\sin^2\theta} = \frac{1}{\sin\theta}$

   $\Rightarrow \frac{x}{-a\sin^2\theta} + \frac{y}{a\sin\theta} = 1$

   Sum of intercepts = $a\left(\sin^2\theta + \sin\theta\right)$ = 2a when $\sin\theta = 1$

10. $f'(x) = \frac{\sin x - x\cos x}{\sin^2 x}$

Let $h(x) = \sin x - x\cos x$
$h'(x) = x\sin x > 0 \ \forall\ 0 < x \le 1$
$h(x)$ is increasing
$h(0) < h(x) \Rightarrow \sin x - x\cos x > 0$ for $0 < x \le 1$
$f'(x) > 0$
Similarly $g(x)$ is decreasing

12. $\int e^x\left(\tan^{-1}x + \frac{1}{1+x^2} - \frac{1}{1+x^2} + \frac{2x}{\left(1+x^2\right)^2}\right)dx = e^x\left(\tan^{-1}x - \frac{1}{1+x^2}\right) + c$

13. $\int\left(x^2+x\right)\left(x^{-8}+2x^{-9}\right)^{1/10}dx = \int(x+1)\left(x^2+2x\right)^{1/10}dx = \frac{5}{11}\left(x^2+2x\right)^{11/10} + c$

14. Value of definite integral can be zero when either the lines coincide or f becomes identically zero for given interval or $f(x) = 0$ has some root in the given interval.
Here $\sin(x + k) = 0$ in $n \in [0, 1]$

17. Let for particular x_0
$g(x_0) = a + k$ then $g(x) = a - k$
$\therefore\ h(x_0) = 2a$
$\therefore\ h(x) = 2a\ \forall\ x \in R$
$\therefore\ h(x)$ is many one and into

18. $y = \frac{x-1}{-x^2+\alpha+1}$; put $y = -t\ \therefore\ t \in \left[\frac{1}{3}, 1\right]$
Hence, we get $\alpha \in \left(-\infty, -\frac{1}{4}\right]$

20. As $n \to \infty$
$\lim\limits_{n\to\infty} f(n) = \lim\limits_{n\to\infty} f(n+2) = \lim\limits_{n\to\infty} f(n+1) = k$ (say)
$\lim\limits_{n\to\infty} f(n+2) = \lim\limits_{n\to\infty} \frac{1}{2}\left\{f(n+1) + \frac{9}{f(n)}\right\}$
$k = \frac{1}{2}\left\{k + \frac{9}{k}\right\} \Rightarrow k = 3$

21. $f(x) = x^2 - \sin x\tan x$
$f'(x) = 2x - \sin x(\sec^2 x + 1)$
$\therefore\ f'(x) < 0 \Rightarrow f(x)$ is decreasing
$f(x) < f(0)$ as $x > 0$
$\therefore\ x^2 - \sin x\cos x < 0$

24. Put separately $x = \frac{1}{2}$, $x = 1$ and $x = 2$ we get
(A) $f(2) + f\left(\frac{1}{2}\right) = 1$ (C) $f(2) + f(1) = f\left(\frac{1}{2}\right)$ (D) $f(2) + f(1) = 0$

28. $f'(x) = 3x^2 - 2x + 100 > 0$

30. $f(1) = f(3)$ ..... (1)
$f' = 3ax^2 + 3bx + 11 = 0$ ..... (2)
Solving we get $a = 1$ and $b = -6$